A COUNTEREXAMPLE AT $n = 164$ TO A CONJECTURED FORMULA FOR THE DENOMINATORS OF PARTIAL SUMS OF RECIPROCAL OCTAGONAL NUMBERS

MATH0

Abstract. Let $a(n)$ be the denominator, in lowest terms, of $V(n) = \sum_{k=0}^{n} 1/((k+1)(3k+1))$ (OEIS A294512), and let $A250400(m)$ be the numerator of the harmonic mean of the first m octagonal numbers. In 2017 W. Lang recorded in the OEIS the conjecture that $a(n) = A250400(n+1)/(n+1)$ for all $n \geq 0$. We show that the conjecture holds at n exactly when $n+1$ is coprime to the numerator of $V(n)$, and that it fails at $n = 164$: there 11 divides both 165 and the numerator of $V(164)$, and $11 \cdot A250400(165) = 165 \cdot a(164)$, so $A250400(165)/165 = a(164)/11$, which is not an integer. To our knowledge this is a new result, although an elementary one: the formula is still marked as a conjecture in the OEIS, and no earlier counterexample has been recorded. The refutation is verified in Lean 4 against a statement that math0 formalized from the OEIS entry, whose definitions are checked by the kernel against the listed OEIS terms; this statement is not part of the formal-conjectures repository.

1. Introduction

The octagonal numbers are $O_j = j(3j-2)$, $j \geq 0$ (OEIS A000567 [6]). In Lang's notation [10], the partial sums of their reciprocals are

$$V(n) = V(3,1;n) = \sum_{k=0}^{n} \frac{1}{(k+1)(3k+1)} = \sum_{j=1}^{n+1} \frac{1}{O_j}, \qquad n \geq 0,$$

and they converge to $(\pi/\sqrt{3} + 3\log 3)/4 = 1.2774\ldots$ [10]. Sums of this kind arise from series $\sum_{k\geq 1} c(k)/k$ with periodic coefficients, a topic going back to Euler and treated in Koecher's book [3, pp. 189–193]; see the comments in [10]. The OEIS sequence A294512, contributed by Wolfdieter Lang on November 1, 2017 [10], lists the denominators $a(n)$ of $V(n)$ in lowest terms, and states that the corresponding numerators are given in A250401 [9], i.e. that $\operatorname{num} V(n) = A250401(n+1)$; we shall see that this identification also fails at $n = 164$ (Corollary 7).

Three years earlier, Colin Barker had contributed the sequences A250400 and A250401 [8, 9]: the numerators and denominators of the harmonic mean $H(m) = m/\sum_{j=1}^{m} O_j^{-1} = m/V(m-1)$ of the first m octagonal numbers, with tables for $m \leq 1000$. Because $H(n+1) = (n+1)/V(n)$, one expects the numerator of $H(n+1)$ to be $(n+1)\,a(n)$. Lang recorded this in the FORMULA section of A294512 as a conjecture, $a(n) = A250400(n+1)/(n+1)$ for $n \geq 0$, and added a comment to A250401 that assumes it ("assuming that $n+1$ divides $A250400(n+1)$"). The formula holds for the 23 listed terms $n \leq 22$, and in fact for all $n \leq 163$.

Result. The conjecture holds at n if and only if $\gcd(n+1, \operatorname{num} V(n)) = 1$ (Lemma 4), and this fails at $n = 164$, where 11 divides 165 and $\operatorname{num} V(164)$ (Lemma 5). The formal proof checks $11 \cdot A250400(165) = 165 \cdot a(164)$ by kernel evaluation and derives a contradiction (Theorem 3). Below 5000 the failures are $n = 164$, 2413, 4520 (Remark 8; not formalized).

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Status and prior work. The conjecture still carries the tag "[conjecture]" in the current revision of A294512 (revision #13, November 12, 2017), and the comment in A250401 is still stated conditionally. The math0 referees read the edit histories of A294512, A250400 and A250401 and checked the sibling entries A294513 and A294515; none records a counterexample. They also found no discussion on the SeqFan mailing list, MathOverflow, Mathematics Stack Exchange or arXiv. The conjecture is not among the OEIS statements of the formal-conjectures repository [2], nor among the 492 OEIS conjectures, formalized in Lean by Tsoukalas et al., that make up the OEIS Open benchmark [1]. We therefore classify the result as *new*, with two caveats. First, it is elementary and of modest significance: the counterexample is found by exact rational arithmetic over 165 terms. Second, the decisive value was implicitly available since 2014: Barker's table of A250400 [8] lists $A250400(165)$, which equals $15\,a(164)$; what was apparently never done is to compare it with the denominator of $V(164)$, since the table of A294512 stops at $n = 22$.

2. Statement

For $x \in \mathbb{Q}$ we write $x = \operatorname{num}(x)/\operatorname{den}(x)$ in lowest terms with $\operatorname{den}(x) \geq 1$.

Definition 1. For $n \geq 0$ and $m \geq 1$ let

$$V(n) = \sum_{k=0}^{n} \frac{1}{(k+1)(3k+1)}, \qquad a(n) = \operatorname{den} V(n),$$

$$H(m) = \frac{m}{\sum_{j=1}^{m} 1/(j(3j-2))}, \qquad A250400(m) = \operatorname{num} H(m).$$

Thus a is A294512 (offset 0) and $A250400$ is A250400 (offset 1). The FORMULA section of A294512 (W. Lang, Nov. 01, 2017) contains, verbatim:

```
a(n) = A250400(n+1)/(n+1), n >= 0.  [conjecture].
```

Conjecture 2 (Lang [10])**.** *For every $n \geq 0$, $A250400(n+1)/(n+1) = a(n)$, the quotient being taken in $\mathbb{Q}$.*

Theorem 3. $11 \cdot A250400(165) = 165 \cdot a(164)$. *Consequently $A250400(165)/165 = a(164)/11 \neq a(164)$, and Conjecture 2 is false.*

In addition, $11 \nmid a(164)$ (Corollary 7), so $A250400(165)/165$ is not even an integer. Since $a(164) \geq 1$, the conjecture also fails at $n = 164$ if the quotient is read as integer division: $\lfloor a(164)/11 \rfloor < a(164)$.

Lean formalization. The statement was formalized by math0 in the file `lean/Math0/Statements/OEIS/A294512.lean`; it was reviewed for fidelity and frozen (SHA-256 `20fd25a6...f89848`) before any proof attempt. Its definitions and the statement are, verbatim except that docstrings and three data lemmas are omitted:

```
namespace Math0.Statements.A294512

def V (n : ℕ) : ℚ := ∑ k ∈ Finset.range (n + 1), 1 / (((k : ℚ) + 1) * (3 * (k : ℚ) + 1))

def a (n : ℕ) : ℕ := (V n).den

def octagonal (j : ℕ) : ℕ := j * (3 * j - 2)

def harmonicMean (m : ℕ) : ℚ := (m : ℚ) / ∑ j ∈ Finset.Icc 1 m, 1 / (octagonal j : ℚ)

def A250400 (m : ℕ) : ℕ := (harmonicMean m).num.natAbs
```

```
def statement : Prop :=
  ∀ n : ℕ, (A250400 (n + 1) : ℚ) / ((n : ℚ) + 1) = (a n : ℚ)

end Math0.Statements.A294512
```

The encoding matches Definition 1. `V` is the rational sum $V(n)$, and `a` is its reduced denominator `Rat.den`, which is positive. `octagonal` is $j(3j-2)$ computed in $\mathbb{N}$; the truncated subtraction $3j-2$ is exact for $j \geq 1$, and only $j \in \{1, \ldots, m\}$ enters `harmonicMean`. `A250400` is the reduced numerator `Rat.num` of the harmonic mean, an integer that is positive for $m \geq 1$, so `Int.natAbs` returns it unchanged. The statement quantifies over all $n \in \mathbb{N}$, i.e. $n \geq 0$, and divides in $\mathbb{Q}$ by $n+1 \neq 0$, so it asserts that $n+1$ divides $A250400(n+1)$ with quotient $a(n)$.

The file also contains three lemmas, each proved by `decide +kernel`, that compare the definitions with the OEIS data: `a_data` checks all 23 listed terms $a(0), \ldots, a(22)$ of A294512 (from $1, 8, 168, 420, \ldots$ to 973748862277067676800), `A250400_data` checks all 20 listed terms $A250400(1), \ldots, A250400(20)$ (from $1, 16, 504, \ldots$ to 162446703896568000), and `octagonal_data` checks $O_0, \ldots, O_4 = 0, 1, 8, 21, 40$ against A000567.

3. Proof

The formal proof has two steps: a kernel evaluation of the identity $11 \cdot A250400(165) = 165 \cdot a(164)$ (Proposition 6), and a short deduction of the negation of the statement from it (Section 3.4). Before giving them we explain why the identity holds; Lemmas 4 and 5 are not part of the formal proof, but together they give a second, hand-checkable proof that the conjecture fails at $n = 164$.

3.1. **Reduction to a coprimality condition.** For $n \geq 0$ write $V(n) = p_n/q_n$ in lowest terms, so $p_n, q_n \geq 1$, $\gcd(p_n, q_n) = 1$ and $q_n = a(n)$, and put $g_n = \gcd(n+1, p_n)$.

Lemma 4. *For every $n \geq 0$ we have $A250400(n+1) = (n+1)\,a(n)/g_n$, that is,*

$$g_n \cdot A250400(n+1) = (n+1) \cdot a(n), \qquad \text{and} \qquad \operatorname{den} H(n+1) = p_n/g_n.$$

In particular Conjecture 2 holds at n if and only if $\gcd(n+1, \operatorname{num} V(n)) = 1$.

Proof. Substituting $j = k+1$ gives $(k+1)(3k+1) = j(3j-2)$, so $\sum_{j=1}^{n+1} 1/O_j = V(n)$ and $H(n+1) = (n+1)/V(n) = (n+1)q_n/p_n$. Since $\gcd(q_n, p_n) = 1$, we have $\gcd((n+1)q_n, p_n) = \gcd(n+1, p_n) = g_n$. So the reduced form of $H(n+1)$ has numerator $(n+1)q_n/g_n$ and denominator p_n/g_n, both positive. Hence $A250400(n+1) = (n+1)q_n/g_n$ and $\operatorname{den} H(n+1) = p_n/g_n$, so $A250400(n+1)/(n+1) = a(n)/g_n$, which equals $a(n)$ if and only if $g_n = 1$. □

3.2. **The prime 11 divides the numerator of $V(164)$.** Let $\mathbb{Z}_{(11)}$ denote the ring of rationals whose reduced denominator is not divisible by 11. For $x, y \in \mathbb{Z}_{(11)}$ and $e \geq 1$ we write $x \equiv y \pmod{11^e}$ if $x - y \in 11^e \mathbb{Z}_{(11)}$.

Lemma 5. $V(164) \in 11\mathbb{Z}_{(11)}$. *Equivalently, $11 \mid p_{164}$; and then $11 \nmid q_{164} = a(164)$.*

Proof. From $\frac{3}{3j-2} - \frac{1}{j} = \frac{2}{j(3j-2)}$ we get $2V(164) = \sum_{j=1}^{165} \left(\frac{3}{3j-2} - \frac{1}{j}\right)$. Since $3 \cdot 8 - 2 = 22$, we have $11 \mid 3j-2$ if and only if $j \equiv 8 \pmod{11}$. Let $J_1 = \{8 + 11t : 0 \leq t \leq 14\}$ be the set of such $j \leq 165$, and let $J_2 = \{11i : 1 \leq i \leq 15\}$ be the set of multiples of 11 up to 165. For $j = 8 + 11t$ we have $3j - 2 = 11(3t+2)$, and $j = 11i$ with $i = t+1$ runs over J_2 as t runs over $0, \ldots, 14$. Hence

$$\sum_{j \in J_1} \frac{3}{3j-2} - \sum_{j \in J_2} \frac{1}{j} = \frac{1}{11} \sum_{t=0}^{14} \left(\frac{3}{3t+2} - \frac{1}{t+1} \right) = \frac{W}{11}, \qquad W = \sum_{t=0}^{14} \frac{1}{(t+1)(3t+2)},$$

and therefore

$$2V(164) = U + \frac{W}{11}, \qquad U = \sum_{\substack{j\le 165\\ j\notin J_1}} \frac{3}{3j-2} - \sum_{\substack{j\le 165\\ j\notin J_2}} \frac{1}{j} \in \mathbb{Z}_{(11)}.$$

The term U. As j runs over $1,\dots,165$, it runs 15 times through a complete residue system modulo 11, and so does $3j-2$, since $j \mapsto 3j-2$ is a bijection of $\mathbb{Z}/11\mathbb{Z}$. Hence

$$U \equiv 45\sum_{r=1}^{10} r^{-1} - 15\sum_{r=1}^{10} r^{-1} = 30\sum_{r=1}^{10} r^{-1} \pmod{11}.$$

As $r \mapsto r^{-1}$ permutes the nonzero residues modulo 11, $\sum_{r=1}^{10} r^{-1} \equiv \sum_{r=1}^{10} r = 55 \equiv 0 \pmod{11}$. Thus $U \in 11\mathbb{Z}_{(11)}$.

The term W. This is Lang's sum $V(3,2;14)$ from the sibling entry A294513 [11]. Its 15 terms have denominators $(t+1)(3t+2) = 2, 10, 24, 44, 70, 102, 140, 184, 234, 290, 352, 420, 494, 574, 660$. Only the terms with $t = 3, 10, 14$ have a denominator divisible by 11, and these three sum to an element of $\mathbb{Z}_{(11)}$:

$$\frac{1}{44} + \frac{1}{352} + \frac{1}{660} = \frac{1}{11}\Big(\frac{1}{4} + \frac{1}{32} + \frac{1}{60}\Big) = \frac{1}{11}\cdot\frac{143}{480} = \frac{13}{480}.$$

So $W \in \mathbb{Z}_{(11)}$. In fact W is divisible by 11^2 in $\mathbb{Z}_{(11)}$: exact summation gives $W = 832564679309/1157486934240$ in lowest terms, where $832564679309 = 11^2 \cdot 31 \cdot 59 \cdot 3762001$ and $1157486934240 = 2^5 \cdot 3^2 \cdot 5 \cdot 7 \cdot 13 \cdot 17 \cdot 19 \cdot 23 \cdot 29 \cdot 41$. (The OEIS lists this numerator and denominator as $A250328(15)$ and $A294513(14)$ [7, 11].) Hence $W/11 \in 11\mathbb{Z}_{(11)}$.

Combining, $2V(164) \in 11\mathbb{Z}_{(11)}$, and since 2 is a unit in $\mathbb{Z}_{(11)}$, $V(164) \in 11\mathbb{Z}_{(11)}$. Writing $V(164) = 11c/d$ with $c, d \in \mathbb{Z}$ and $11 \nmid d$, we get $p_{164}d = 11cq_{164}$, so $11 \mid p_{164}$; and $11 \nmid q_{164}$ because $\gcd(p_{164}, q_{164}) = 1$. □

Lemmas 4 and 5 already refute Conjecture 2: $g_{164} = \gcd(165, p_{164})$ is divisible by 11, so $A250400(165)/165 = a(164)/g_{164}$ differs from $a(164)$. To find g_{164} exactly one needs p_{164} modulo 3 and 5.

3.3. **The certificate.** Exact rational arithmetic gives $V(164) = p/q$ with $p = p_{164}$ and $q = q_{164} = a(164)$ equal to the following 157-digit integers (each printed on two lines):

$p =$ 194229398903971676694086965665455427842637116406947138890743624836635802968683411350986133152586219273131422041236122282490381187601638511468776846769541861 9,

$q =$ 152290093400122011048693127705500110226862752969298710633405928964347790649504483184038053251578276692377828360641749997086983953873238253689047621338899200 0.

Here $p \equiv 1 \pmod 3$, $p \equiv 4 \pmod 5$, $p \equiv 0 \pmod{11}$, $p \not\equiv 0 \pmod{121}$, and $q \equiv 5 \pmod{11}$.

Proposition 6. *$A250400(165) \cdot 11 = 165 \cdot a(164)$.*

Proof. Formal proof. This is an equality of two closed natural-number expressions. In Lean it is the private theorem `key`, proved by `decide +kernel`: the kernel evaluates both sides directly from the definitions of Section 2 and confirms that they coincide (Section 4).

Explanation. By the residues above, $g_{164} = \gcd(3 \cdot 5 \cdot 11, p) = 11$, so Lemma 4 gives $11 \cdot A250400(165) = 165 \cdot a(164)$. □

Corollary 7. *$A250400(165) = 15\,a(164)$, $\operatorname{den} H(165) = \operatorname{num} V(164)/11$, and $11 \nmid a(164)$. In particular $A250400(165)/165 = a(164)/11 \notin \mathbb{Z}$.*

Proof. The first equality is Proposition 6 divided by 11; the second is Lemma 4 with $g_{164} = 11$; the third is Lemma 5. □

The value $A250400(165) = 15q$, a 158-digit number, agrees with Barker's table [8] for $m \le 1000$, which two math0 referees checked independently against Definition 1 for all $m \le 1000$. By the second equality, $A250401(165) = p/11$ rather than p, so the comment of Lang in A250401 that identifies $A250401(n+1)$ with the numerator of $V(n)$ also fails at $n = 164$, as its stated assumption does; so does the comment in A294512 that the corresponding numerators are given in A250401.

3.4. **Proof of Theorem 3.** This follows the Lean proof of `result` step by step. Suppose Conjecture 2 holds, and take $n = 164$: $A250400(165)/165 = a(164)$ in $\mathbb{Q}$. Multiplying by $165 \neq 0$ gives $A250400(165) = 165 \cdot a(164)$ in $\mathbb{Q}$, and hence in $\mathbb{N}$, because the map $\mathbb{N} \to \mathbb{Q}$ is injective. By Proposition 6, $11 \cdot 165 \cdot a(164) = 11 \cdot A250400(165) = 165 \cdot a(164)$, so $1650 \cdot a(164) = 0$ and $a(164) = 0$. But $a(164) = \operatorname{den} V(164) \ge 1$. This contradiction shows that Conjecture 2 is false. The relation $A250400(165)/165 = a(164)/11$ follows from Proposition 6 by dividing by $11 \cdot 165$, and it differs from $a(164)$ because $a(164) \neq 0$. □

Remark 8. The following facts come from exact rational computations in Python and are not formally verified. For $0 \le n < 5000$, $\gcd(n+1, \operatorname{num} V(n)) > 1$ exactly for $n = 164$, 2413 and 4520, with gcd 11, 17 and 137; so $n = 164$ is the least counterexample. Lang's analogous conjecture in A294513 [11], for $V(3,2;n) = \sum_{k \le n} 1/((k+1)(3k+2))$ and pentagonal numbers, holds for $n < 2500$ except at $n = 1891$, where 11 divides both 1892 and $\operatorname{num} V(3,2;1891)$.

4. Machine verification

Files and declarations. The proof is in the file `lean/Math0/Frontier/OEIS/A294512.lean` of the math0 repository, `https://github.com/Foundation-0/math0`. It imports the frozen statement file `lean/Math0/Statements/OEIS/A294512.lean` (Section 2) and the audit commands in `lean/Math0/Audit.lean`. Its contents after the module docstring are, verbatim except that docstrings are omitted:

```
namespace Math0.Frontier.A294512
open Math0.Statements.A294512

set_option maxHeartbeats 4000000 in
private theorem key : A250400 165 * 11 = 165 * a 164 := by
  decide +kernel

theorem result : ¬ statement := by
  intro h
  have h1 := h 164
  have hk := key
  have hpos : 0 < a 164 := (V 164).den_pos
  have h2 : (A250400 165 : ℚ) = 165 * (a 164 : ℚ) := by
    rw [div_eq_iff (by norm_num)] at h1
    push_cast at h1 ⊢
    linarith
  have h3 : A250400 165 = 165 * a 164 := by exact_mod_cast h2
  omega

end Math0.Frontier.A294512

#math0_settles Math0.Frontier.A294512.result Math0.Statements.A294512.statement
```

The main declaration `Math0.Frontier.A294512.result` has type `¬Math0.Statements.A294512.statement`; its proof is that of Section 3.4, with `omega` doing the final arithmetic.

Kernel computation. The lemma `key` is the certificate of Proposition 6. With `decide +kernel`, the Lean kernel itself reduces the decision procedure for this equality in $\mathbb{N}$ to `true`: it unfolds the definitions of Section 2, adds the 165 rationals of each of the two sums, normalizing to lowest terms with its built-in arbitrary-precision arithmetic on natural-number literals, and compares the two resulting 159-digit numbers. No compiled code is trusted; `native_decide`, which would add the axiom `Lean.ofReduceBool`, is not used. The option `maxHeartbeats 4000000` applies to `key` only. The whole file compiles in about 20 seconds on our machine, most of it spent loading Mathlib.

Audit. The file ends with `#math0_settles`, defined in `lean/Math0/Audit.lean` for statements that math0 formalizes itself. It fails unless (1) `statement` is a definition of type `Prop` without universe parameters; (2) `statement` and `result` depend on no axioms other than `propext`, `Classical.choice` and `Quot.sound` (this rejects `sorryAx`, `Lean.ofReduceBool`, `Lean.trustCompiler` and custom axioms); and (3) the type of `result` is syntactically the statement or its negation, after removing metadata annotations, with no unfolding. On this file it reports

```
math0 audit: 'Math0.Statements.A294512.statement' depends on axioms [Classical.choice, Quot.
    sound, propext]
math0 audit: 'Math0.Frontier.A294512.result' depends on axioms [Classical.choice, Quot.sound,
    propext]
math0 statement check: 'Math0.Frontier.A294512.result' refutes 'Math0.Statements.A294512.
    statement'
```

Toolchain and reproduction. We use Lean 4 [5] `v4.33.1` and Mathlib [4] `v4.33.1`, pinned in `lean/`; the repository also pins formal-conjectures [2] at commit `c77fdd4`, which this result does not use. To reproduce, run `scripts/verify.sh` from the repository root: it fetches the Mathlib build cache, builds all math0 results with `lake build` (re-running every audit command), checks that the pinned `formal_conjectures` and `mathlib` checkouts are unmodified, and reports the number of passed statement checks. The single file can be checked with `lake env lean Math0/Frontier/OEIS/A294512.lean` in the directory `lean`.

References

[1] T. Adamczewski, *OEIS Open: How many conjectures can language models turn into theorems?*, arXiv:2608.11941, 2026.

[2] M. Firsching, P. Lezeau, S. Mercuri, M. Z. Horváth, Y. Dillies, C. Sönne, E. Wieser, F. Zhang, T. Hubert, B. Agüera y Arcas, and P. Kohli, *Formal Conjectures: An open and evolving benchmark for verified discovery in mathematics*, arXiv:2605.13171, 2026. Repository: `https://github.com/google-deepmind/formal-conjectures`.

[3] M. Koecher, *Klassische elementare Analysis*, Birkhäuser, Basel, Boston, 1987.

[4] The mathlib Community, *The Lean mathematical library*, in: Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020), ACM, 2020, 367–381.

[5] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, in: Automated Deduction – CADE 28, Lecture Notes in Comput. Sci. **12699**, Springer, 2021, 625–635.

[6] The OEIS Foundation Inc., `https://oeis.org/A000567` (octagonal numbers).

[7] The OEIS Foundation Inc., `https://oeis.org/A250328`, entry by C. Barker (2014).

[8] The OEIS Foundation Inc., `https://oeis.org/A250400`, entry by C. Barker (2014).

[9] The OEIS Foundation Inc., `https://oeis.org/A250401`, entry by C. Barker (2014), comment by W. Lang (2017).

[10] The OEIS Foundation Inc., `https://oeis.org/A294512`, entry by W. Lang (2017).

[11] The OEIS Foundation Inc., `https://oeis.org/A294513`, entry by W. Lang (2017).